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A Deep Learning Approach to Automate Classification of Arsenic-Affected Skin using EfficientNet-B1

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***Abstract*—Arsenic contamination in groundwater is a major health hazard, particularly in areas where it causes chronic arsenic poisoning, which is manifested in skin lesions and other dermatological conditions. Early and accurate detection of arsenic-affected skin is important for timely medical intervention. This paper proposes the automatic classification system using EfficientNet-B1, which is one of the most outstanding deep learning models for detecting arsenic-induced skin lesions using clinical images, in the model, training, and testing of a dataset comprising healthy and arsenic-affected skin, with augmentation techniques employed to increase robustness and generalizability from the image dataset. Several deep earning models were compared for this classification task: VGG-19, ResNet-50, MobileNet-v2, EfficientNet-B0, and EfficientNetB1. EfficientNet-B1, after fine-tuning, achieved the highest performance of 94.69% accuracy and proved outstanding in this application. These results point out that EfficientNet-B1 is a reliable and scalable tool for automatic diagnosis and can thus assist healthcare providers in resource-constrained settings. Future work will include an extension of the dataset, improving model interpretability, and exploring clinical practical use of the system. .**

***Index Terms*—Deep learning, EffficientNet-B1, arsenic-affected skin, automated diagnosis, medical imaging.**

I. Introduction

Arsenic contamination in groundwater is a significant environmental and public health threat, especially in Bangladesh, India, and Southeast Asia, where millions of people are in danger of chronic arsenic poisoning. Excess arsenic exposure can lead to skin sores, rough skin (hyperkeratosis), and an increased risk of cancer [14] [7]. The presence of melanosis and keratosis, along with high arsenic concentrations in body tissues like nails, hair, and skin scales serve as key indicators of arsenical dermatosis. Therefore, arsenicosis can be detected through skin images of arsenic-affected areas [15]. Diagnosis of arsenic-affected skin through clinical knowledge and visual inspection is subjective, time-consuming, and susceptible to human error [13]. Automated and objective procedures are essential for improving diagnosis accuracy and accessibility, especially in resource-limited situations.

A deep learning approach like Convolutional Neural Networks (CNNs) has shown potential in diagnosing skin cancer, diabetic retinopathy, and pneumonia using X-ray pictures. [21] [4] [5]. However, there is an insufficient study on automatic classification of arsenic-affected skin. This work uses EfficientNet-B1, a cutting-edge deep learning model that balances accuracy and computational efficiency, making it ideal for deployment in clinical and low-resource applications.

This research utilizes the ArsenicSkinImageBD dataset, a comprehensive collection of high-resolution images depicting both healthy and arsenic-affected skin of individuals from arsenic-endemic regions [3]. The dataset contains a wide range of images showing different stages of skin problems caused by arsenic. This makes it an ideal resource for training robust classification models. By employing data augmentation techniques and transfer learning, the EfficientNet-B1 model is fine-tuned to accurately distinguish between healthy and affected skin, aiming to enhance diagnostic capabilities.

II. Research Objective

A. Main Objective

The main goal of this research is to evaluate the accuracy of the EfficientNet B1 model in classifying skin as either infected or not infected by arsenic, along with a comparative analysis of the results from other similar models using the same dataset.

B. Research Question

This study aims to address the following research questions:

1) How efficiently does the EfficientNet-B1 model detect arsenic-affected skin lesions using the dataset?
2) How does the EfficientNet-B1 model compare to other deep-learning models for classifying arsenic-affected skin?
3) Can the proposed model achieve a balance between accuracy and computational efficiency suitable for deployment in resource-limited settings?
4) How do data augmentation and transfer learning impact the model's ability to generalize to unseen arsenic-affected skin images?

III. Related Works

Deep learning and dermatology—an exciting intersection during recent years. It gets more interesting when skin diseases take into account those caused by even environmental toxins like arsenic. The prospect of classifying skin lesions that are affected by arsenic through an advanced neural network architecture such as EfficientNet-B1 offers a promising direction toward improving diagnostic precision and efficiency. This section reviews the relevant literature, underlining the importance of the deep learning methodology in the dermatological application related to arsenic-induced skin conditions.

Chronic exposure to arsenic is known to cause a range of skin disorders, including thickened skin (hyperkeratosis), darkened patches (hyperpigmentation), and serious conditions like Bowen's disease and squamous cell carcinoma [9] [19]. The development of these conditions is complex, involving factors like oxidative stress and disruptions in the immune system, which have been widely reported in scientific studies [20]. For example, research by Huang et al. [8] highlights how arsenic exposure increases reactive oxygen species and triggers abnormal immune responses, which together contribute to the development of cancer. Understanding these mechanisms is essential for creating reliable diagnostic tools that can effectively identify and classify the different types of arsenic-induced skin lesions.

The application of deep learning in dermatology has shown remarkable promise, with studies demonstrating that convolutional neural networks (CNNs) can achieve diagnostic performance comparable to that of experienced dermatologists. Marchetti et al. trained a CNN on a large dataset of dermoscopic images, achieving dermatologist-level accuracy in skin cancer classification [12]. Similarly, Boffetta et al. conducted a systematic review that highlighted the effectiveness of machine learning algorithms in identifying skin lesions associated with arsenic exposure [1]. These findings suggest that deep learning models can be trained to recognize subtle features in skin lesions that may be indicative of arsenic toxicity. Afterward, by early care and intervention, these risks can be minimized.

Moreover, the integration of deep learning with image processing techniques has been explored to enhance the classification of skin lesions. Long et al. proposed a novel lightweight deep learning framework that leverages segmentation approaches to improve classification accuracy [6]. This methodology is particularly relevant for arsenic-affected skin lesions, as the visual characteristics of these lesions can vary significantly among individuals. The ability of deep learning models to adapt to these variations is crucial for developing robust diagnostic tools.

Besides the progress in deep learning models, genetic factors have also been explored in the context of arsenic-related skin issues. Research by Luo et al. looked at how genetic variations, known as gene polymorphisms, are linked to the development of skin lesions in people exposed to high levels of arsenic [11]. This study highlights the importance of considering genetic predispositions when designing machine learning models for classifying skin lesions. Including genetic data in these models could improve their accuracy and make diagnoses more tailored to individual patients, leading to more personalized healthcare.

Furthermore, the ethical considerations surrounding the collection of skin samples from arsenic-exposed individuals pose challenges for traditional diagnostic methods. Yang et al. highlighted the difficulties in obtaining skin samples for research purposes, which necessitates the development of non-invasive diagnostic techniques [19]. Deep learning approaches, particularly those utilizing image data, can circumvent these ethical issues by relying on readily available clinical images for training and validation.

The importance of comprehensive datasets in training deep learning models cannot be overstated. The performance of these models is heavily dependent on the quality and diversity of the training data. Recent studies have emphasized the need for large, annotated datasets that encompass a wide range of skin lesions, including those caused by arsenic exposure [2] [18]. By utilizing diverse datasets, researchers can ensure that their models are capable of generalizing across different populations and skin types, thereby improving diagnostic accuracy.

In conclusion, the integration of deep learning techniques in the classification of arsenic-affected skin lesions holds significant promise for advancing dermatological diagnostics. The existing literature provides a robust foundation for understanding the complexities of arsenic-induced skin disorders and the potential of machine learning to enhance diagnostic accuracy. Future research should focus on expanding datasets, incorporating genetic factors, and addressing ethical challenges to fully realize the potential of deep learning in this critical area of public health.

IV. Methodology

This section outlines the methodology employed in the development and evaluation of the proposed automated classification system for arsenic-affected skin using the EfficientNet-B1 model. It is structured into several key phases: dataset preparation, model architecture selection, training and tuning, and evaluation of performance metrics. Each phase is crucial for ensuring the robustness and accuracy of the classification model.

A. Data Collection and Preparation

In the experiment, the ArsenicSkinImageBD dataset was employed, consisting of 8,892 skin images. The dataset was further augmented for improvement and thus increased to a total number of 10,180 skin images. The dataset is evenly split, with half images depicting skin affected by arsenic exposure and half images showing mixed skin conditions that are not affected by arsenic.

This balanced dataset is essential for mitigating bias during the training process and ensuring that the model learns to differentiate effectively between the two classes. The images

Fig. 1. The palm of an affected person on the left and the palm of a healthy person on the right.

Fig. 2. The skin of an affected person on the left and the skin of a healthy person on the right.

are diverse, capturing various stages and types of arsenic-induced dermatological manifestations, along with healthy and other non-arsenic-related skin conditions.

1) Data Preprocessing: Before training the model, the images were resized to 240×240 pixels to fit the input requirements of the EfficientNet-B1 architecture. Image normalization was performed to scale pixel values between 0 and 1, enhancing the training stability.

2) Data Augmentation: To improve the model's robustness and generalizability, data augmentation techniques such as random rotations, flips, shifts, and zooms were applied to the dataset, and this added 1288 more images to the dataset. This approach helps to simulate variability in real-world clinical settings and prevents overfitting.

3) Dataset Split: The dataset was split into two subsets: 70% for training and 30% for validation and testing. The split was performed randomly to ensure a balanced representation of arsenic-affected and healthy skin images in each subset.

B. Training and Tuning

We fine-tuned five EfficientNet-B1, EfficientNet-B0, MobileNetV2, ResNet50, and VGG19 pre-trained models using our dataset and followed the transfer learning approach. The training process consists of splitting the dataset into 70% for the training set and the remaining 30% for validation and test sets.

During training, the batch size was 32 with 20 epochs. It utilized the Adam optimizer, which can handle sparse gradients and adapt the learning rate in a dynamic way. The loss function used here is the categorical cross-entropy, which is suitable for multi-class classification problems. A dynamic learning rate schedule was adopted based on the validation loss, and early stopping was implemented to avoid overfitting.

EfficientNet-B1 and EfficientNet-B0 use compound scaling to achieve a good balance between the different aspects of model accuracy and computational efficiency. MobileNetV2 introduced inverted residual blocks with depthwise convolutions for lightweight computations. ResNet50 uses residual connections to avoid vanishing gradients in backpropagation so that deep networks can be trained, while VGG19, a 19-layer deep network architecture, learns hierarchical features. This setup had very robust training across the board for all models.

C. Model Selection and Proposed Model Architecture

The EfficientNet-B1 architecture was chosen for this study due to its state-of-the-art performance in image classification tasks, particularly in the medical domain [16]. EfficientNet models are known for their efficiency and accuracy, achieved through a compound scaling method that uniformly scales the depth, width, and resolution of the network. This model performs exceptionally well with small datasets and those with limited variation, making it less prone to overfitting.

The key factor in EfficientNet is the compound scaling method, which balances the network's depth, width, and input resolution. Instead of independently scaling these dimensions, EfficientNet uses the compound scaling approach, which scales them together in a way that maximizes performance while minimizing computational cost.

EfficientNet uses a compound scaling method that balances the network's depth, width, and resolution. This allows for better performance with fewer parameters compared to traditional CNN architectures.

$$\text{depth} = \alpha^d, \quad \text{width} = \beta^d, \quad \text{resolution} = \gamma^d \tag{1}$$

$$\alpha \times \beta^2 \times \gamma^2 \approx 2 \tag{2}$$

EfficientNet-B1 is constructed with Mobile Inverted Bottleneck Convolution (MBConv) blocks, using depthwise separable convolutions to save the number of parameters. It also uses Squeeze-and-Excitation (SE) blocks for recalibration of features that further help in improvement [16].

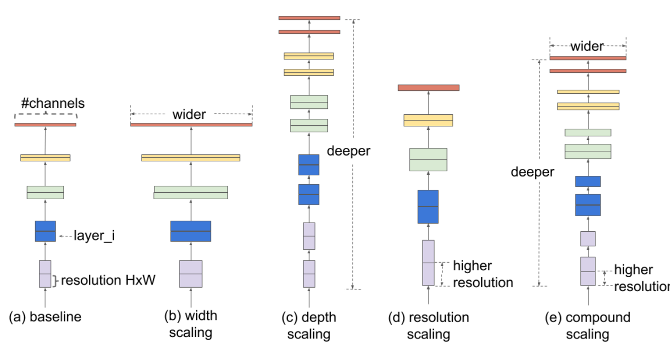

Fig. 3. EfficientNet Architecture

Swish is applied as an activation function in this network for better flow of gradients and increased accuracy. So, this allows the model to perform better when having fewer parameters than that of classic CNN architectures. It is pre-trained on ImageNet, takes 240×240 pixel resolution images as input, and has a dropout rate of 0.2 to reduce overfitting. The architecture ensures high accuracy with a low computational cost [16].

D. Evaluation Metrics

Accuracy is the main evaluation metric here, along with other performance metrics such as precision, recall, and F1-score, which were calculated to provide a comprehensive evaluation of the model's performance. These metrics are particularly important in medical image classification, where the cost of false negatives can be significant. A confusion matrix was also generated to visualize the model's performance across the two classes, providing insights into areas where the model may need improvement.

E. Implementation Details

The model was implemented using Python with TensorFlow and Keras on an NVIDIA GPU T4-enabled environment, significantly accelerating the training process. This study employed multiple pre-trained deep learning models, including EfficientNet-B1, VGG19, MobileNet-V2, ResNet-50, and EfficientNet-B0, to classify arsenic-affected skin lesions. Each model was fine-tuned and trained on a well-prepared dataset with augmentation techniques to enhance generalizability. The combination of a well-prepared dataset, robust model architecture, and thorough training and evaluation processes contributed to the high accuracy achieved, highlighting the potential of deep learning in dermatological diagnostics.

V. Results

This section presents the performance evaluation of the EfficientNet-B1 model and compares it with other models working on the same dataset.

The results show that EfficientNet-B1 outran others with the best train accuracy of 96.86% and test accuracy of 94.69% while demonstrating excellent generalization. ResNet-50 followed this with strong performance: 90.37% training and 85.88% testing accuracy, while EfficientNet-B0 showed 91.61% for training and 87.28% for testing—a fine trade-off for efficiency and accuracy. MobileNetV2, optimized for resource constraints, delivered 89.30% training and 84.32% testing accuracy, while VGG19 had the lowest performance, achieving 86.97% training and 79.65% testing accuracy. Overall, EfficientNet-B1 proved the most effective for detecting arsenic-induced skin lesions.

TABLE I
Training and Testing Accuracy of Different Models

Model	Training Accuracy (%)	Testing Accuracy (%)
VGG-19	86.97	79.65
ResNet-50	90.37	85.88
MobileNetV2	89.30	84.32
EfficientNet-B0	91.61	87.28
EfficientNet-B1	96.86	94.69

The comparison between the models based on their training and testing accuracy can be shown in a chart as in Figure 4.

For the selected model, EfficientNet-B1, more evaluation metrics are involved to describe outcomes.

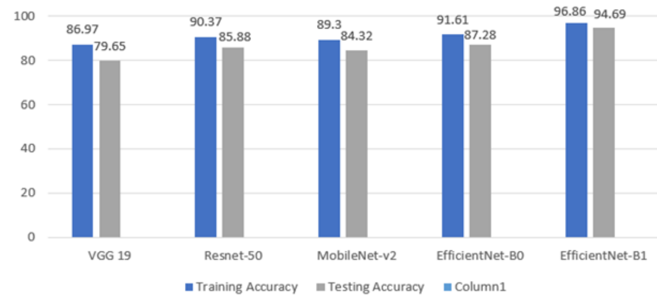

Fig. 4. Different Models' Result Chart

TABLE II
Performance Metrics of the EfficientNet-B1 Model

Metric	Value
Accuracy	0.9469
Precision	0.9430
Recall	0.9545
F1 Score	0.9480
AUC (Area Under the Curve)	0.98
True Positives (TP)	1507
False Positives (FP)	91
True Negatives (TN)	1385
False Negatives (FN)	71

A. Performance Metrics

The classification performance of the EfficientNet-B1 model was evaluated using the following metrics:

1) Accuracy: The model was able to correctly classify almost all test images with an accuracy of 94.69%, hence showing its efficiency in distinguishing affected skin from non-affected skin.

2) Precision: The precision score of the model was 94.3%, which perfectly identified arsenic-affected cases without false positives.

3) Recall: The recall score was 95.45%, indicating that the model detected almost all the images of arsenic-affected skin.

4) F1 Score: The F1 score of 94.8% reflects the model's balanced performance in effectively minimizing both false positives and false negatives.

B. Confusion Matrix

The confusion matrix, as shown in Figure 5, provides a detailed breakdown of the model's predictions:

1) True Positives (TP): 1507 images of arsenic-affected skin were correctly classified as "infected."

2) True Negatives (TN): 1385 images of non-affected skin were correctly classified as "Not Infected."

3) False Positives (FP): 91 images were incorrectly classified as "infected" when they were actually "not infected."

4) False Negatives (FN): 71 images were incorrectly classified as "Not Infected" when it was actually "Infected."

C. ROC Curve and AUC Analysis

The ROC curve is a plot of the trade-off between true positive rate (sensitivity) and false positive rate across thresholds.

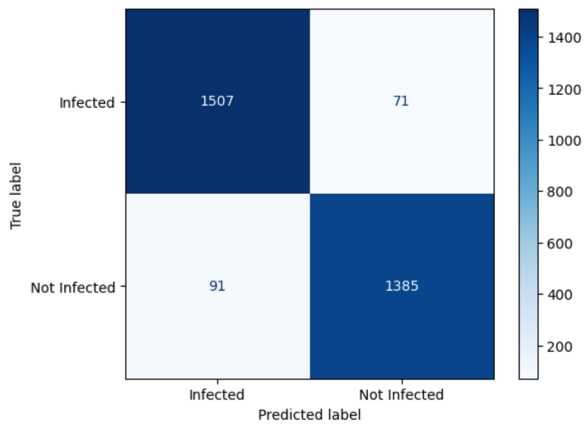

Fig. 5. Confusion Matrix

It had an AUC of 0.98, while the model performed well in differentiating arsenic-affected skin images from non-affected skin images. The formula used to calculate AUC is:

$$\text{AUC} = \int_0^1 \text{TPR}(t), d(\text{FPR}(t)) \tag{3}$$

This high AUC confirms the reliability and robustness of the model with minimum false negatives and false positives, suitable for clinical applications.

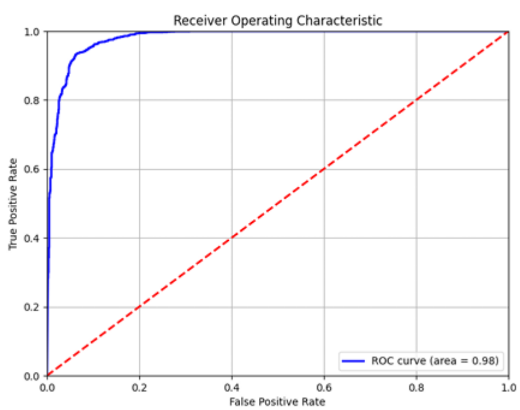

Fig. 6. ROC-Curve

D. Implications of the Study

The application of the EfficientNet-B1 model is highly promising in clinical use, especially in areas with high exposure to arsenic. Automated skin classification systems are quick, inexpensive, and accurate for preliminary diagnosis, thus being particularly suitable for rural and resource-constrained areas with limited access to dermatologists. This technology can also be adapted into mobile applications that help people self-screen for arsenic-induced skin conditions and seek medical advice on time. Besides, decision-support deployment of such models would lighten the workload of medical experts and ensure that more patients get timely evaluations. This innovation is in line with public health efforts to try to reduce the impact of environmental hazards such as arsenic exposure.

E. Limitations

Despite the promising results, the study has some limitations that should be considered:

1) Dataset Constraints: The model was trained and evaluated using the ArsenicSkinImageBD dataset, which, although comprehensive, may not fully capture the global variability in arsenic-affected skin conditions. Variations in lighting, image quality, and skin types across different populations could affect the model's generalizability.

2) Lack of External Validation: The model was not tested on external datasets, which limits the ability to validate its performance in different real-world settings. Future studies should involve cross-dataset validation to assess the robustness and adaptability of the model across diverse populations.

3) False Negatives: Although the recall score is high, some cases of arsenic-affected skin may still be missed, which could impact the clinical applicability of the model. Further refinement is needed to ensure that no critical cases are overlooked, especially in a medical context.

F. Future Directions

Future research could address the limitations and focus on the following areas to improve the model's performance and broaden its applicability:

1) Cross-Dataset Validation: To assess the model's robustness and generalizability, cross-dataset validation should be carried out using more diverse and larger datasets from different populations and environments.

2) Model Interpretability: The interpretability of the model is an important feature for clinical applications. Techniques like Grad-CAM can help in understanding the model's decision-making process, thus gaining trust and usability among clinicians.

3) Integration of Clinical Data: Incorporating additional clinical data, such as demographic information and medical history, could improve the model's accuracy and help in creating a more comprehensive diagnostic tool.

4) Expanding the Dataset: Incorporating additional images from diverse populations and environmental conditions could improve the model's ability to generalize across different settings. Data augmentation techniques that simulate real-world variability could also be beneficial.

5) Improving Recall: Techniques such as adjusting the decision threshold, employing ensemble models, or fine-tuning class weights could be explored to improve recall without significantly affecting precision.

VI. Conclusion

This study has proved the capability of the EfficientNet-B1 deep learning model in classifying arsenic-affected skin lesions autonomously. Trained on the augmented Arsenic-SkinImageBD dataset, expanded to 10,180 images, the model achieved a remarkable testing accuracy of 94.69% and F1 score of 94.8%, which are strong results and compare well to other deep learning models used for similar image classification tasks [10].

Comparative performance analysis against other models such as VGG-19, ResNet-50, MobileNetV2, and EfficientNet-B0 proved that EfficientNet-B1 had outperformed them in training and testing accuracy, thus proving to be the best performer for this task.

Going beyond mere classification, the proposed system can help medical experts reach a diagnosis and intervene with ailments caused by arsenic to attain better outcomes. In general, this work serves as a pointer to new paths advanced machine learning techniques offer to dermatological diagnostics [17].

However, generalization of the results cannot be assured with only one data set. Future studies will be required to validate the model with larger and more diverse data sets and integrate clinical data to enhance accuracy and interpretability.

This study provides evidence of the power that deep learning has in medical image analysis, even with pathologies due to environmental exposure, such as arsenic, and opened newer avenues toward developing more robust diagnostic tools.

Acknowledgment

We would like to express our gratitude to the developers of the ArsenicSkinImageBD dataset, which was instrumental in this research. This comprehensive dataset provided a valuable resource for training and evaluating the deep learning model for classifying arsenic-affected skin conditions. The dataset is accessible at ArsenicSkinImageBD [3].

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